a Instrument detection



Video frame x_t

640 px input setting

Backbone

Multiscale fusion

Detection heads

P_3

P_4

P_5

C

C

C

C

C = concatenation

box

class



Boxes, classes, scores

$s \geq 10^{-3}$ ≤ 100 boxes

b Question-driven template answering

What type of forceps. . . ?

Question q

$r(q)$

Grammar router

polar

purpose

entity

Affirmative

Fixed purpose

Entity prior e_f^*

Training tool labels

Compose $T_{r(q)}$

The type of forceps. . .

Declarative response $\widehat{a}(q)$